

Fashion Store Sales & Customer Analytics using SQL

Introduction

This project focuses on analyzing a fashion retail dataset using SQL to derive meaningful business insights. The database simulates real-world e-commerce operations, including customer data, product details, sales transactions, and inventory management.

The goal is to transform raw data into actionable insights that support data-driven decision-making.

■ Objectives

- Analyze sales performance across countries and channels
- Identify top-selling products and categories
- Perform customer segmentation based on spending
- Track monthly sales trends and growth
- Evaluate inventory and stock levels
- Understand customer purchasing behavior

■ Tools & Technologies

- **MySQL Workbench** – Database creation & querying
- **SQL** – Data analysis
- **Excel (optional)** – Data preprocessing
- **Power BI (optional)** – Visualization

■ Database Design

Tables Used:

- Customers
- Products
- Sales
- SalesItems
- Campaigns
- Channels
- Stock

■ Key Concepts:

- Primary Keys (PK)
- Foreign Keys (FK)
- Relational Database Design
- Normalization

■ Data Modeling

The database follows a **relational schema** where:

- Customers → linked to Sales
- Sales → linked to SalesItems
- SalesItems → linked to Products
- Products → linked to Stock

This ensures:

- Data integrity
- Reduced redundancy
- Efficient querying

■ Key SQL Operations Performed

◆ Joins

- Combined multiple tables to extract meaningful insights

◆ Aggregations

- SUM(), AVG(), COUNT() used for KPIs

◆ Subqueries

- Used for nested analysis

◆ Window Functions

- RANK(), DENSE_RANK(), ROW_NUMBER(), LAG()

-- Key Analysis & Queries

1. Revenue Analysis

- Total revenue by country
- Channel-wise performance

2. Customer Analysis

- Customer Lifetime Value (CLV)
- High-value customers
- Repeat customers

3. Product Analysis

- Top-selling products
- Revenue by category

- Product contribution

4. Time-Based Analysis

- Monthly sales trend
- Month-over-month growth

5. Inventory Analysis

- Low stock products
- Stock vs sales comparison

■ Sample Advanced Query

-- Top-selling product in each category

```
SELECT *
FROM (
  SELECT
    category,
    product_name,
    total_sold,
    DENSE_RANK() OVER (
      PARTITION BY category
      ORDER BY total_sold DESC
    ) AS rnk
  FROM (
    SELECT
      p.category,
      p.product_name,
      SUM(si.quantity) AS total_sold
    FROM salesitems si
    JOIN products p
      ON si.product_id = p.product_id
    GROUP BY p.category, p.product_name
  ) t1
```

) t2

WHERE rnk = 1;

■ Key Insights

- Certain products consistently generate higher revenue
- Online channel contributes significantly to total sales
- High-value customers drive majority of revenue
- Seasonal trends impact sales performance
- Some products show low stock but high demand

■ Challenges Faced

- Handling SQL errors (1055, 1064)
- Managing GROUP BY with strict mode
- Combining aggregation with window functions
- Ensuring proper joins across multiple tables

■ Solutions Implemented

- Used subqueries to resolve aggregation conflicts
- Applied proper GROUP BY techniques
- Optimized queries for performance
- Structured database using relational design

■ Outcome

The project successfully demonstrates the ability to:

- Work with relational databases
- Perform advanced SQL analysis
- Extract actionable business insights
- Solve real-world data problems

■ Conclusion

This project highlights strong proficiency in SQL and data analysis by transforming raw retail data into meaningful insights. It showcases the ability to handle real-world datasets, perform complex queries, and support business decision-making.